

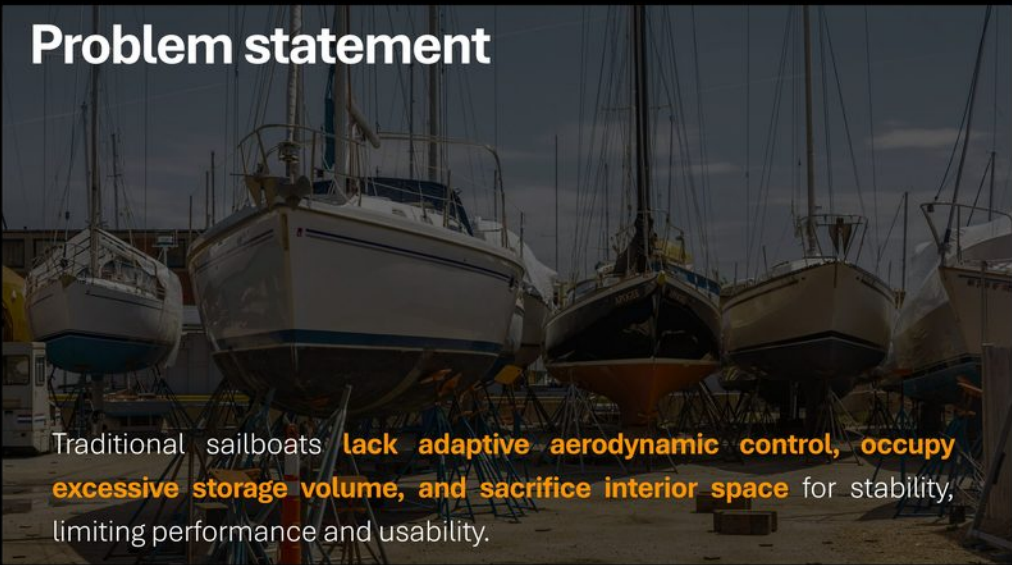
CHIROPTERA: THE BIO-INSPIRED SAILBOAT



A bio-inspired leisure sailboat combining **adaptive bat-wing sail mechanics with a streamlined whale-inspired hull for greater efficiency, agility, and spatial economy.**

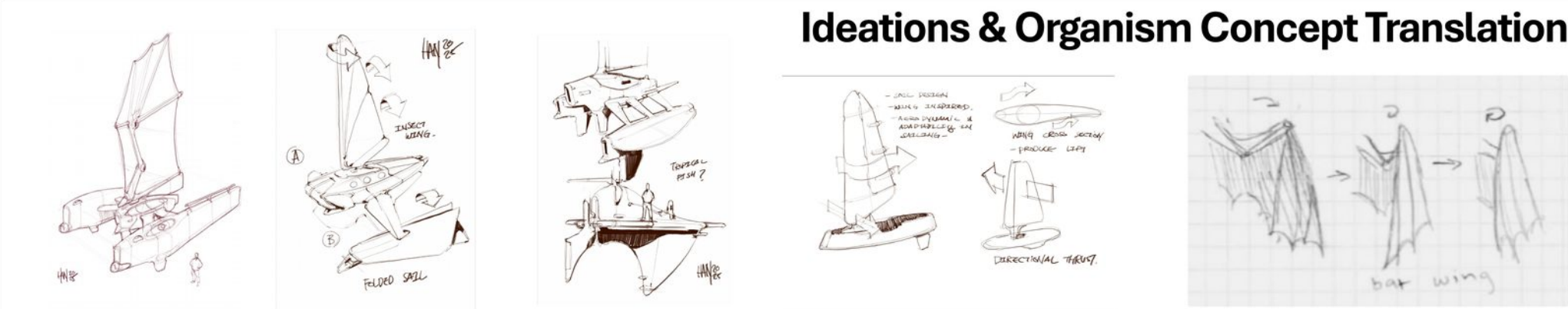
Fall 2025 Yasha Jeltuhin, Han Chi Chang, Brigitte Flannery, Maya Angia, Alex Cordovado BIOS-4740-A Instructor: Dr. Jeanette Yen

Problem statement



Traditional sailboats **lack adaptive aerodynamic control, occupy excessive storage volume, and sacrifice interior space** for stability, limiting performance and usability.

Ideations & Organism Concept Translation



The challenge was to identify biological systems whose innate geometries—flexible wings, streamlined bodies, or compact folding mechanisms—could **translate into improved marine performance.**

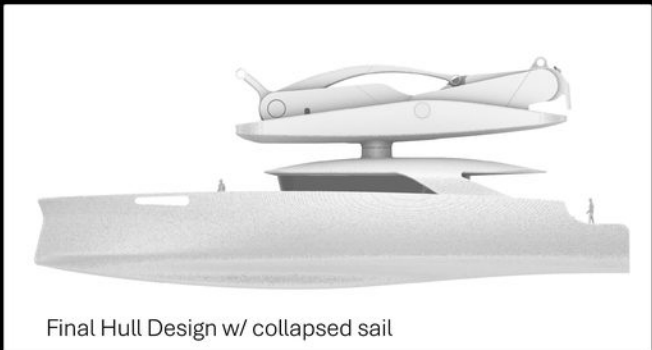
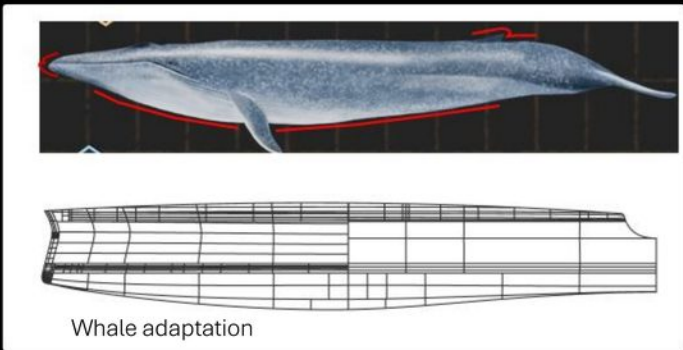
Proposed Sail Structure



Modeled after the multi-joint flexibility of a bat wing, **the sail structure integrates articulated supports and a tensioned membrane system that fold seamlessly into the boom and expand into an optimized aerodynamic profile on command.**

Proposed Hull Structure

Inspired by the tapered geometry and fluid surface continuity of whales, **the hull integrates broad mid-section volume with refined shaping to reduce drag while creating a more spacious, luxury-oriented interior.**

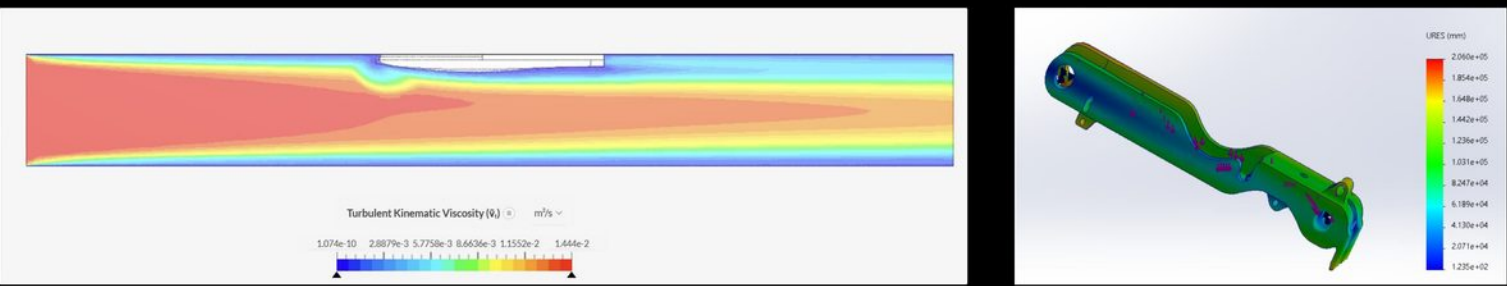


Prototype Testing



Scaled water and wind tests validated the sail's adaptive geometry and the hull's hydrodynamic stability, **showing measurable gains in speed, maneuverability, and flow efficiency across multiple prototype iterations.**

Computational Validation



Computational validation through CFD and FEA confirmed **aerodynamic efficiency, structural reliability, and improved flow behavior**, guiding key refinements to both the sail system and hull geometry.

In-Scene Renderings



Renderings demonstrate the boat's form, performance, and visual presence across realistic sailing environments.

Structural Translation

